INCH-POUND

MIL-P-70944(AR)

30 July 1990

MILITARY SPECIFICATION

PANEL ASSEMBLY, FRONT: 12922129

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Front Panel Assembly, used in the M90 Radar Chronograph (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-P-14232	- Parts, Equipment and Tools for Army Materiel, Packaging of
MIL-F-23419/9	- Fuse, Cartridge, Instrument Type, Style FM09M (Nonindication)
MIL-C-26482	- Connector, Electrical (Circular, Miniature, Quick Disconnect, Environment Resisting) Receptacles And Plugs, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1290

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MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-2164 - Environmental Stress Screening Process for Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12922105 - EMI Test Procedure
12922119 - Front Panel, Schematic Diagram
12922129 - Front Panel, Electronic Assembly
12922225 - Environmental Test Methods

SPECIAL PACKAGING INSTRUCTIONS

P12922129 - Special Packaging Instruction for the Front Panel Assembly

(Copies of drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in

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this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General description. The Front Panel Assembly is an interface unit used to control the M90 Radar Chronograph and to display measured projectile velocity.

3.2 Materials. The materials shall be in accordance with the applicable drawings and specifications.

3.3 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 12922129, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Panel Assembly shall meet the requirements of MIL-F-13926 and MIL-STD-454 as follows :

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship
e. Soldering electrical connections

3.4 Performance. The Panel Assembly contains three connectors with rubber waterproof covers, two D-type connectors, three rotary switches, one toggle switch, two pushbutton switches, two light indicators, one Light Emitting Diode display and one fuse.

3.4.1 Power. Toggle switch shall be designated as S1. When in ON position the following pins of connectors J1 and J4 shall be connected: J1-A, J1-E, J4-3, J4-14, J4-15, J4-16 and J4-26.

3.4.2 Fuse. Protection fuse - 3 Amp, 250 V in accordance with MIL-F-23419/9.

3.4.3 Interconnections. The interconnections shall be in accordance with Drawing 12922119.

3.4.3.1 Input power. Input power connector shall be designated as J1. Connector type MS3124E-14-5P of MIL-C-26482, shall be located on the front side of the Panel Assembly.

3.4.3.2 Antenna interconnect. Connector type MS3124E-14-5SY of MIL-C-26482, shall be located on the front side of the Panel Assembly, designated as J2.

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3.4.3.3 Data output. Connector type MS3124E-22-55S of MIL-C-26482, shall be located on the front side of the Panel Assembly, designated as J3.

3.4.3.4 Connector J4. Connector type M24308/2-5 of MIL-C-26482 , shall be located on the rear side of the Panel Assembly.

3.4.3.5 Connector J5. Connector type M24308/2-15 of MIL-C-26482, shall be located on the rear side of the Panel Assembly.

3.4.4 Caliber. A 4 position rotary switch shall be designated as S2, and shall be located on the front side of the Panel Assembly. At each position, the following pins of J3 and J5 shall be connected to panel ground in accordance with TABLE I.

TABLE I - Caliber switch operation

SWITCH POSITION	DESIGNATION ON THE PANEL	GROUNDED PINS
1	105	NONE
2	155	J5-10,J3-m
3	175	J5-43,J3-X
4	203	J5-43,J5-10, J3-X ,J3-m

3.4.5 Expected velocity. A 12 position rotary switch shall be designated as S3, and shall be located on the front side of the Panel Assembly. At each position, the following pins of J3 and J5 shall be connected to panel ground in accordance with TABLE II.

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TABLE II - Expected velocity switch operation

SWITCH POSITION	DESIGNATION ON THE PANEL	GROUNDED PINS
1	DATA UNIT TEST	NONE
2	150-300	J5-40,J3-z
3	300-400	J5-5,J3-GG
4	400-500	J5-40,J5-5, J3-z, J3-GG
5	500-600	J5-38,J3-HH
6	600-700	J5-38,J3-HH J5-40,J3-z
7	700-800	J5-38,J3-HH J5-5,J3-GG
8	800-900	J5-38,J5-5,J5-40 J3-HH,J3-GG, J3-z
9	900-1000	J5-4,J3-CC
10	1000-1100	J5-4,J3-CC J5-40,J3-z
11	1100-1250	J5-4,J3-CC J5-5,J3-GG
12	SYSTEM TEST	J5-4,J5-5,J5-40 J3-CC, J3-GG, J3-z

3.4.6 Data display. 9 positions rotary switch shall be designated as S4, and shall be located on the front side of the Panel Assembly. At each position, the following pins of J3 and J5 shall be connected to panel ground in accordance with TABLE III.

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TABLE III - Data display switch operation

SWITCH POSITION	DESIGNATION ON THE PANEL	GROUNDED PINS
1	MV	NONE
2	MV1	J5-39,J3-S
3	MV2	J5-6,J3-T
4	MV3	J5-39,J5-8, J3-S, J3-T
5	MV4	J5-7, J3-U
6	MV5	J5-7, J3-U J5-39,J3-S
7	MV6	J5-7, J3-U J5-6, J3-T
8	MV7	J5-7,J5-6,J5-39 J3-U,J3-T,J3-S
9	AVG	J5-41,J3-E

3.4.7 Press before fire (PBF). Pushbutton switch shall be designated S5, and shall be located on the front side of the Panel Assembly. When released, the normally-closed contacts shall connect the panel ground to J5-65 and to J3-y. When pressed shall connect the panel ground to J5-29 and to J3-x.

3.4.8 Clear memory. Pushbutton switch shall be designated as S6, and shall be located on the front side of the Panel Assembly. When pressed shall connect the panel ground to J5-44 and to J3-H.

3.4.9 Standby. Yellow indicator lamp shall be designated as DS1 and shall be rated 28 VDC maximum and 40mA maximum. DS1-1 shall be connected to J5-32 and J3-Z. DS1-2 shall be connected to J4-10.

3.4.10 Ready. Yellow indicator lamp shall be designated as DS2 and shall be rated 28 VDC maximum and 40mA maximum. DS2-1 shall be connected to J5-69 and J3-Y. DS2-2 shall be connected to J4-10.

3.4.11 Display. The display shall consist of six Display Elements, designated from left to right as U1, U2, U3, U4, U5 and U6. The DISPLAY shall have the following characteristics:

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3.4.11.1 Display power. The power consumption shall be: 4.5 to 7 VDC, 1.02 AMP max. The DISPLAY UNIT ground shall be connected to the chassis. The positive supply shall be connected to J4-27.

3.4.11.2 Display data. Four bit binary coded data shall be connected to the Panel Assembly as follows:

D3 : J3-e ,J5-47 (most significant)
D2 : J3-d ,J5-13
D1 : J3-c ,J5-2
D0 : J3-I ,J5-36 (least significant)

3.4.11.3 U1 blanking. Blanking signal to U1 shall be connected to J3-r, J5-37, and shall be active high.

3.4.11.4 Display latch. Each Display Element shall have a LATCH input, activated at the high to low transition. The connections shall be as follows:

Display Element	Latch input pins
U1	J3-w, J5-52
U2	J3-f, J5-14
U3	J3-K, J5-50
U4	J3-M, J5-15
U5	J3-k, J5-3
U6	J3-J, J5-8

3.4.11.5 Hexadecimal display. The left Display Element U1 shall be a hexadecimal digit, with the truth table in accordance with TABLE IV. D0,D1,D2,D3 as in 3.4.11.2.

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TABLE IV - Truth table for hexadecimal display element

BLANK	D3	D2	D1	D0	DISPLAYED
L	L	L	L	L	0
L	L	L	L	H	1
L	L	L	H	L	2
L	L	L	H	H	3
L	L	H	L	L	4
L	L	H	L	H	5
L	L	H	H	L	6
L	L	H	H	H	7
L	H	L	L	L	8
L	H	L	L	H	9
L	H	L	H	L	A
L	H	L	H	H	B
L	H	H	L	L	C
L	H	H	L	H	D
L	H	H	H	L	E
L	H	H	H	H	F
H	X	X	X	X	BLANK

NOTES: L - Logic low level
H - Logic high level
x - Don't care

3.4.11.6 Numeric display. The 5 right Display Elements U2 to U6 shall be numeric digits, that represent a decimal number consisting of four digits for the integer portion of the number and one digit for the fraction. A constantly illuminated decimal point at the right of U5, shall separate the integer and fractional portions of the number. The truth table given in TABLE V describes the operation of each Display Element U2 through U6 .

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TABLE V - Truth table for numeric display elements

D3	D2	D1	D0	DISPLAYED
L	L	L	L	0
L	L	L	H	1
L	L	H	L	2
L	L	H	H	3
L	H	L	L	4
L	H	L	H	5
L	H	H	L	6
L	H	H	H	7
H	L	L	L	8
H	L	L	H	9

NOTES: L - Logic low level
H - Logic high level

3.4.12 Harness. A harness shall provide the interconnections required in 3.4.1 to 3.4.11 and in accordance with the requirements of Drawing 12922119.

3.5 Environmental requirements.

3.5.1 High temperature. The Front Panel Assembly shall meet the requirements of 3.3 and 3.4 when thermally stabilized at +125 degrees F. The Front Panel Assembly when subjected to storage temperature of +150 degrees F shall meet the requirements of 3.3 and 3.4 when returned and thermally stabilized at ambient temperature.

3.5.2 Low temperature. The Front Panel Assembly shall meet the requirements of 3.3 and 3.4 when thermally stabilized at -50 degrees F and when returned and thermally stabilized at ambient temperature.

3.5.3 Solar Radiation. The Front Panel Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to 355 BTU per square foot.

3.5.4 Humidity. The Front Panel Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to 240 hours at a relative humidity of 94 +/-5 percent from +68 degrees F to +125 degrees F.

3.5.5 Vibration. The Front Panel Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to vibration, having a double amplitude of 1.0 inch, from 5 to 6.5 HZ and acceleration level of 2 G's over the 6.5 to 200 HZ frequency

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range for a 15-minute duration in each of its three mutually perpendicular axes.

3.5.6 Altitude. The Front Panel Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to an altitude of 10,000 feet in operational mode and after exposure at an altitude of 50,000 feet in nonoperational mode.

3.5.7 Rain. The Front Panel Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to a 2.0 inch per hour, 20 mile per hour wind driven rain.

3.5.8 Salt fog. The Front Panel Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to 48 hours of continuous exposure to a 5 percent salt solution with a specific gravity of 1.027 to 1.034, a PH of 6.5 to 7.0 and a temperature of 95 degrees F.

3.5.9 Sand and dust. The Front Panel Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to 36 hours of exposure to wind driven sand and dust.

3.5.10 Fungus. The Front Panel Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to 28-day fungus exposure to a composite spore suspension.

3.5.11 Environmental stress screening (ESS) temperature. The Front Panel Assembly shall meet the performance requirements in 3.3 and 3.4 after being subjected to 40 hours of temperature cycling between 0 degrees F and +125 degrees F in accordance with 5.1.2.1 of MIL-STD-2164.

3.5.12 Environmental stress screening (ESS) vibration. The Front Panel Assembly shall meet the performance requirements in 3.3 to 3.4 after being subjected to random motion in the range of 20 to 2000 Hz for 5 minutes at an axis perpendicular to the largest surfaces of the Front Panel Assembly. Acceleration spectral density is 6.06 g. RMS in accordance with 5.1.1 of MIL-STD-2164.

3.5.13 EMI. The Front Panel Assembly shall meet the requirement of Drawing 12922105 when integrated into an M90 RADAR CHRONOGRAPH (NSN 1290-01-073-0746) supplied as Government Furnish Material.

3.6 Workmanship. The workmanship requirements of MIL-F-13926 and MIL-STD-454 shall apply.

3.7 Fabrication. The Panel Assembly shall be manufactured in accordance with Drawings 12922119 and 12922129.

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3.8 First article inspection. When specified in the contract or purchase order, three samples shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5 and all applicable drawings. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 and MIL-STD-454 form a part of the Quality Assurance Provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

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a.	Temperature	+60 degrees to +90 degrees F.
b.	Relative humidity	90% RH maximum.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three Front Panel Assembly for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in TABLE VI and TABLE VII herein, and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any Front Panel Assembly to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this

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specification and Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in TABLE VI shall be performed as noted under the characteristics.

TABLE VI. Functional tests
(100% Inspection)

No.	Characteristics	Requirement	Test Procedure
Critical:	None	None	N/A
Major:			
101	Performance	3.4	4.8
102	ESS Temperature	3.5.11	4.7.1.1
103	ESS Vibration	3.5.12	4.7.1.2
Minor:			
201	Materials	3.2	Applicable Drawing/ Visual
202	Workmanship	3.6	Applicable Drawing/ Visual
203	Fabrication	3.7	Applicable Drawing/ Visual

4.5 Special sampling. Three Front Panel Assemblies shall be selected at random as a special sample from each lot or each 150 produced whichever occurs first. The sample shall meet the requirements and tests of TABLE VII.

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TABLE VII. Environmental sampling tests.

No.	Characteristics	Requirement	Test Procedure
Critical:	None	None	N/A
Major:			
301	High temperature	3.5.1	4.7.3.1
302	Low temperature	3.5.2	4.7.3.2
303	Solar Radiation	3.5.3	4.7.3.3
304	Humidity	3.5.4	4.7.3.4
305	Vibration	3.5.5	4.7.3.5
306	Altitude	3.5.6	4.7.3.6
307	Rain	3.5.7	4.7.3.7
308	Salt Fog	3.5.8	4.7.3.8
309	Sand and Dust	3.5.9	4.7.3.9
310	Fungus	3.5.10	4.7.3.10
311	EMI	3.5.13	4.7.3.11
Minor:	None	None	N/A

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.4.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and

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permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.4.

4.6.2.3 Inspection equipment design requirements.

4.6.2.3.1 Temperature testing. The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5 degrees F (1.4 degrees C). The temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5 degrees F per foot in any direction. The heat source of the test chamber shall be located so that radiant heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber within the supply airstream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.6.2.3.2 Fabrication, setup and calibration. Design and fabrication of test equipment shall be such that it incorporates the features of rigidity, stability, and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instruction shall be prepared and submitted to the procuring activity.

4.7 Methods of inspections. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the government procuring agency for approval prior to use.

4.7.1 Environmental stress screening (ESS). Prior to conducting acceptance testing of this specification, the Front Panel Assembly shall be subjected to 40 hours of run-in-time. The run-in shall exclude any downtime necessary to repairs. It shall include vibration, followed by temperature cycling.

4.7.1.1 ESS temperature. Testing equipment utilized in this test shall conform to the requirements of 4.6.2.3.1. The Front Panel Assembly shall be placed in the chamber and subjected to a 40-hour run-in period which includes vibration and operation at temperature extremes. The temperature shall be raised gradually

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at a rate not to exceed 2 degrees F per minute until thermally stabilized at +125 degrees F for 4 hours duration. Then the temperature chamber shall be decreased to 0 degrees F at the same rate for another 4 hours. Subsequent to the above temperature extremes (+125 degrees F and 0 degrees F), the Front Panel Assembly shall not have been damaged or adversely affected and shall meet the performance requirements period of this specification for the entire 40-hour run-in period.

4.7.1.2 ESS vibration. The Front Panel Assembly in an operating mode shall be exposed to a 5 minute burst of random vibration. Failures occurring during this five minute test shall be accrued(if possible) and corrected at the conclusion of the 5 minute period in accordance with the requirements of 3.3.

The random vibration spectrum shall be:

20 - 80 Hz at 3 dB/octave rise
80 - 350 Hz at 0.04 g sq/Hz
350-2000 Hz at 3 dB/octave rolloff

The motion shall be applied perpendicular to the largest surfaces of the Front Panel Assembly.

4.7.3 Environmental methods of inspection.

4.7.3.1 High temperature. This test shall be conducted in accordance with Drawing 12922225. Subsequent to the exposure, the Front Panel Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.1 of this specification.

4.7.3.2 Low temperature. This test shall be conducted in accordance with Drawing 12922225. Subsequent to the exposure, the Front Panel Assembly shall not be damaged or adversely affected and shall meet the requirements of 3.5.2.

4.7.3.3 Solar radiation. This test shall be conducted in accordance with Drawing 12922225. Subsequent to the exposure, the Front Panel Assembly shall not be damaged or adversely affected and shall meet the requirements of 3.5.3.

4.7.3.4 Humidity. The Front Panel Assembly shall be subjected to the humidity test as outlined in Drawing 12922225. Performance test to be conducted during the exposure shall be conducted during the first,third and fifth cycles; subsequently, the Front Panel Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.4 of this specification.

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4.7.3.5 Vibration. The Front Panel Assembly shall be subjected to the vibration as specified in Drawing 12922225. The displacement and frequency shall be as follows:

GRAVITY	FREQUENCY (Hz)
1.0 inch Double Amplitude	5 - 6.5
2G Acceleration	6.5 - 200

Subsequent to the vibration test, the Front Panel Assembly shall not have been damaged or adversely affected and shall meet the performance requirements in 3.5.5 of this specification. Duration shall be one sweep of 15 minutes in each plane.

4.7.3.6 Altitude. The Front Panel Assembly shall be subjected to altitude test as specified in Drawing 12922225. Subsequent to the exposure, the Front Panel Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.6 of this specification.

4.7.3.7 Rain. The Front Panel Assembly shall be subjected to rain test as specified in Drawing 12922225. Subsequent to the exposure, the Front Panel Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.7 of this specification.

4.7.3.8 Salt fog. The Front Panel Assembly shall be subjected to the salt test as specified in Drawing 12922225. Subsequent to the exposure, the Front Panel Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.8 of this specification.

4.7.3.9 Sand and dust. The Front Panel Assembly shall be subjected to the sand and dust test as specified in Drawing 12922225. Subsequent to the exposure, the Front Panel Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.9 of this specification.

4.7.3.10 Fungus. The Front Panel Assembly shall be subjected to the fungus test as specified in Drawing 12922225, for 28 days. Subsequent to the exposure, the Front Panel Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.10 of this specification.

4.7.3.11 EMI. The Front Panel Assembly shall be subjected to the EMI test procedure in accordance with Drawing 12922105 when installed into an M90 RADAR CHRONOGRAPH.

4.8 Performance. The performance parameters of 3.4.1 through 3.4.12 shall be demonstrated either manually or using a

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contractor designed, Government-approved, automatic test fixture simulating the conditions and making the measurements of providing a visual indication of a PASS or FAIL condition. The performance test shall verify that each of the following functional characteristics conform to the requirements of 3.4.1 through 3.4.12 inclusive.

4.8.1 Power. The ON/OFF switch S1 shall be checked by turning ON and OFF the READY and STANDBY indicators.

4.8.2 Fuse. When removed, the fuse shall open the electrical circuit.

4.8.3 Interconnections. All interconnections between the connectors J1, J2, J3, J4, J5 shall be inspected in accordance with the requirements of drawing 12922119.

4.8.4 Caliber. Switch S2 shall be checked in accordance with TABLE I of 3.4.4 .

4.8.5 Expected velocity. Switch S3 shall be checked in accordance with TABLE II of 3.4.5 .

4.8.6 Data display. Switch S4 shall be checked in accordance with TABLE III of 3.4.6 .

4.8.7 Press before fire. Switch S5 shall be checked in accordance with 3.4.7 .

4.8.8 Clear memory. Switch S6 shall connect, when pressed, the panel ground to J5-44 and to J3-H.

4.8.9 Standby. When a power source of 24 +/- 0.5 Volt is applied to pins J3-Z or J5-32 and J4-10, the STANDBY indicator shall light.

4.8.10 Ready. When a power source of 24 +/- 0.5 Volt is applied to pins J3-Y or J5-69 and J4-10, the READY indicator shall light.

4.8.11 Hexadecimal display. When logical levels are applied to the binary data inputs:

D3 : J3-e ,J5-47
D2 : J3-d ,J5-13
D1 : J3-c ,J5-2
D0 : J3-I ,J5-36
U1 blanking: J3-r, J5-37 - Logic level low.

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The display shall operate in accordance with TABLE IV of 3.4.11.5 for the Hexadecimal Display Element U1.

4.8.12 Numeric display. The operation of 5 numeric Display Elements U2-U6 shall be verified by displaying 8888.8 and 1234.5. The data to U2 through U6 shall be applied in accordance with TABLE V of 3.4.11.6 on binary data inputs:

D3 : J3-e ,J5-47
D2 : J3-d ,J5-13
D1 : J3-c ,J5-2
D0 : J3-I ,J5-36

and using the DISPLAY LATCH for each numeric display in accordance with 3.4.11.4, each at the time.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing and marking shall be in accordance with MIL-P-14232 and Special Packaging Instruction P12922129. The level of protection shall be specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Front Panel Assembly is intended for use in the M90 Radar Chronograph.

6.2 Acquisition requirements. Procurement documents should specify the following :

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Applicable stock number.
d. First Article Sample requirements (see 3.8).
e. Level of preservation and packing (see 5.1).
f. Serialization requirements, if applicable.
g. Certificate of conformance for each lot or shipment of product.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific

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acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirement of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DOD Form 1423.

Reference Paragraph	DID Title	DID No.	Suggested Tailoring
4.6	Quality Engineering Acceptance Inspection Equipment Description Documentation	DI-R-1714	None

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 or as stated in the contract will apply.

6.6 Definitions.

6.6.1 Contractor. Unless otherwise specified, the word Contractor used throughout this specification shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government Owned-Contractor Operated (GOCO), and Government Owned-Government Operated (GOGO) plants.

6.6.2 Technical data package. A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable lists, specifications, standards, models, performance requirements, quality assurance provisions and packaging data.

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6.7 Government furnished property. The contracting officer should arrange to furnish the property listed in 3.5.13.

6.8 Subject term (key word) listing.

M90 Radar Chronograph (M90)
Panel Assembly, Front (Panel)

Custodian:
Army-AR

Preparing activity:
Army-AR

(project 1290-A456)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70944 (AR)	30 July 1990

3. DOCUMENT TITLE

PANEL ASSEMBLY, FRONT: 12922129

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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